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A novel atomic *J*-integral concept beyond conventional fracture mechanics

Pengfei Jia [a], Kai Huang [a,b,c,*], Hongjun Yu [a], Takahiro Shimada [d], Licheng Guo [a], Takayuki Kitamura [e]

[a] Department of Astronautic Science and Mechanics, Harbin Institute of Technology, Harbin 150001, China
[b] Institute for Advanced Ceramics, Harbin Institute of Technology, Harbin 150001, China
[c] State Key Laboratory for Strength and Vibration of Mechanical Structures, Xi'an Jiaotong University, Xi'an 710049, China
[d] Department of Mechanical Engineering and Science, Kyoto University, Nishikyo-ku, Kyoto 615-8540, Japan
[e] Kyoto University, Sakyo-ku, Kyoto 606-8501, Japan

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ABSTRACT

The *J*-integral, which is crucial to the nonlinear fracture analysis within the framework of continuum fracture mechanics, fails at the atomic scale due to the atomic discreteness emerging with decreasing size. In this paper, a novel atomic *J*-integral calculation method is proposed for atomically sharp cracks by developing a new integral scheme to describe the local displacement gradient and stress field for discrete models. The applicability of the atomic *J*-integral is verified by comparing with traditional FEM and stress-based method, and its path-independency is elucidated by evaluating the geometric nonlinearity at the atomic scale. Furthermore, it is corroborated that the atomic *J*-integral method is dimension-independent even at the ultra-small scale below the dimensional limit of continuum mechanics.

1. Introduction

How to determine the growth of preexisting cracks in materials is the paramount topic in fracture mechanics. Generally, the cracking in brittle materials is dominated by linear elasticity at the macroscale, since the nonlinear deformation is highly localized in the vicinity of the crack tip. As the material size shrinks down to the nanoscale, however, such atomic-level nonlinearity plays a crucial role in the fracture process and should be fully considered [31]. Eshelby's seminal work [5] about the configuration force proposed a conservative integral as a mathematical tool, until Rice [26] originally introduced the *J*-integral concept and clarified its physical significance. For nonlinear fracture mechanics (NLFM), the *J*-integral evaluates the energetic force driving the propagation of an existing crack and the intensity of the crack tip fields in nonlinear elastic continuum [32,33]. The nonlinear elastic fracture refers to the pure brittle fracture of the material without any plastic behavior. The path independence also provides great convenience for solving the *J*-integral by numerical simulations, considering that the measurement of the energy released at the crack tip is difficult. As the critical value J_C refers to the fracture toughness, the *J*-integral criterion possesses a rigorous theoretical basis and physical significance.

Since the *J*-integral was constructed within the framework of continuum fracture mechanics (CFM), it remains an open question whether the *J*-integral works for cracking issues in nano-sized structures or not. When the crack dimension decreases to the nanoscale, the singular stress field, which intrinsically dominates the fracture in the theory of CFM, will shrink down to a few nanometers. In such cases, only an extremely smaller number of atoms exist near the crack tip, compared with that in the macroscale materials. Therefore, the postulate that a large number of atoms exist in the vicinity of the crack tip to regard the area as a continuum is no longer valid at the atomic scale. And this leads to the breakdown of CFM [46,30,9,6,7,35,43]. According to a series of studies about the failure of continuum theories at the nanoscale, various criteria within the framework of CFM, including the Griffith energy release rate (ERR) criterion [30], *J*-integral criterion [31] and strength criterion [9,15], will break down when describing the brittle elastic fracture at the nanoscale.

To break through the lower dimensional limit of CFM, some attempts have been made by applying the atomic fracture mechanics (AFM). A representative example is the atomic ERR method, which characterizes the first bond breaking at the crack tip [30,9,10,31,15]. Nevertheless, it is still not easy for the atomic ERR method to deal with multiscale fracture problems, e.g., an extremely small (nano-sized) crack in a large (macro-sized) component. Because the atomic ERR is calculated by counting the change of strain energy of the whole structure. Considering

* Corresponding author at: Department of Astronautic Science and Mechanics, Harbin Institute of Technology, Harbin 150001, China.
E-mail address: huangkai@hit.edu.cn (K. Huang).

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Nomenclature

a	Length of crack
a_0	Lattice constant of sp^2 C—C bond
$\boldsymbol{a}_i^e$	Displacement vectors of atoms
d_c	Critical deformation at fracture
E	Young's modulus
E_{ij}	Green strain
$f_j^{\alpha\beta}$	Interatomic force
H	Length of monolayer specimen
$\mathbf{I}$	Identity matrix
J	J-integral
J_{AFM}	AFM-based J-integral
J_{crk}	J-integral along the crack face
J_{FEM}	J-integral calculated by FEM
J_{MS}	J-integral calculated by stress-based method
J_{C}	Critical value of J/fracture toughness
J_{AFM}^{F}	Critical value of J_{AFM}
$J_{\mathrm{FEM}}^{\mathrm{F}}$	Critical value of J_{FEM}
n_i	Outward normal vector
N_i^e	Shape function
q	Weight function
$r_i^{\alpha\beta}$	Distance between atoms
u_i	Displacement
$\boldsymbol{u}$	Displacement vector of Gaussian points
V_0	Volume per atom
W	Strain energy density
α_y	Yield offset
$\varepsilon^{\mathrm{global}}$	Global strain
δ_{ij}	Kronecker delta
Γ_0	Closed contour around the crack tip
Γ_{B}	Integral path
Γ_{C}	Crack face
σ^{global}	Global stress
σ_{ij}^{α}	Atomic stress
$\boldsymbol{\sigma}$	Stress vectors of Gaussian points
$\boldsymbol{\sigma}_i^e$	Stress vectors of atoms
AFM	Atomic fracture mechanics
AIREBO	Adaptive intermolecular reactive empirical bond order
CFM	Continuum fracture mechanics
EDI	Equivalent domain integral
ERR	Energy release rate
FEM	Finite element method
GNR	Graphene nanoribbon
LEFM	Linear elastic fracture mechanics
LAMMPS	Large-scale atomic/molecular massively parallel simulator
MS	Molecular static
NLFM	Nonlinear elastic fracture mechanics

that the J-integral only focuses on the local area near the crack tip, it may provide a new approach to solving such kinds of issues. However, there remains a fundamental question that whether the criterion constructed at the macroscale still works at the nanoscale due to the extremely confined singular stress field at the nano-sized crack tip [45].

Some efforts have been put to develop insight into the application of the J-integral at the atomic scale using atomistic simulation methods [42,3,19,17,27,18,1,11,12] first studied this issue by calculating an approximation for the J-integral, i.e., the H-sum, by the atomic potential energy and an estimate of the displacement gradients of designated atoms around the crack tip. Nakatani et al. [20,21] avoided the difficulties of calculating a contour integral by applying an equivalent domain integral (EDI) approach. In theirwork, the contribution of each atom to the J-integral is weighted spatially according to each atom's coordinates. Then, Jin et al. [16] also used the EDI approach to calculate the atomic J-integral, but applied a different way to compute the strain energy density. Jones et al. [17] applied a Lagrangian kernel-based estimator of continuum fields to atomic data in order to estimate the J-integral for the analysis of cracks and dislocations. Roy et al. [28] investigated the validity of linear elastic fracture mechanics (LEFM) to model fracture processes at the discrete atomic scale and implemented the atomic J-integral as a nanoscale fracture metric to investigate flaw-tolerance at the nanoscale. However, these existing studies for the atomic J-integral are mainly based on the domain integral form calculated by the contribution of designated atoms in the vicinity of the crack tip. The displacement gradient is hard to be described accurately for a discrete atomic structure. Moreover, it is still unclear whether the atomic J-integral method is path-independent or dimensional-independent when the material size decreases below the dimensional limit of CFM.

In this paper, an AFM-based J-integral (J_{AFM}) concept is proposed to solve the nonlinear fracture of nanostructures. We avoid calculating the displacement gradients of atoms by meshing the atomic structure with quadrilateral isoparametric elements and adding Gaussian points inside the elements. The line integral along the crack face is considered in the J-integral. The accuracy and path-independency are verified by a series of numerical examples based on the graphene nanoribbons (GNRs). Furthermore, we demonstrate the breakdown of the conventional J-integral based on NLFM and give the lower dimensional limit at the atomic scale. The dimensional-independency of J_{AFM} introduced in this paper is discussed by comparing it with the CFM method.

2. Atomic J-integral

According to our previous work, both the Griffith ERR and the stress intensity factors with in the framework of LEFM are applicable when the specimen is lager than the lower dimensional limits of LEFM [30]. However, when the specimen dimension is further reduced, the influence brought by the material nonlinearity will become much stronger. Therefore, when describing the fracture of nanostructures below the critical scale of LEFM, the strong nonlinearity and existence of few atoms near the crack tip must be fully considered. According to NLFM established on continuum fracture theory, J-integral is the most commonly used parameter for characterizing nonlinear fracture behavior. As shown in Fig. 1a, the local coordinates at the crack tip are introduced in a 2D atomic system, where the atoms and bonds are presented by dots and solid lines. The crack surfaces are modeled by removing an array of atomic bonds and marked by red dots. The basic form of J-integral is defined as [26]

$$J = \lim_{\Gamma_\varepsilon \to 0} \int_{\Gamma_\varepsilon} \left(W\delta_{1i} - \sigma_{ij} u_{j,1} \right) n_i d\Gamma \tag{1}$$

where Γ_ε is an infinitesimal integral path around the crack tip; n_i is the outward normal vector of Γ_ε, as shown in Fig. 1a; δ_{ij} is Kronecker delta and subscripts i, j take the values of 1 and 2; the repetition of an index in a term denotes a summation with respect to that index over its range; a comma denotes a partial derivative; W is the strain energy density defined as $W = \int_0^{\varepsilon_{ij}} \sigma_{ij} d\varepsilon_{ij}$.

In practical numerical calculations, J-integral in contour integral form is numerically inaccurate and inconvenient to compute for arbitrary finite element mesh configurations. Thus, Eq. (1) is usually

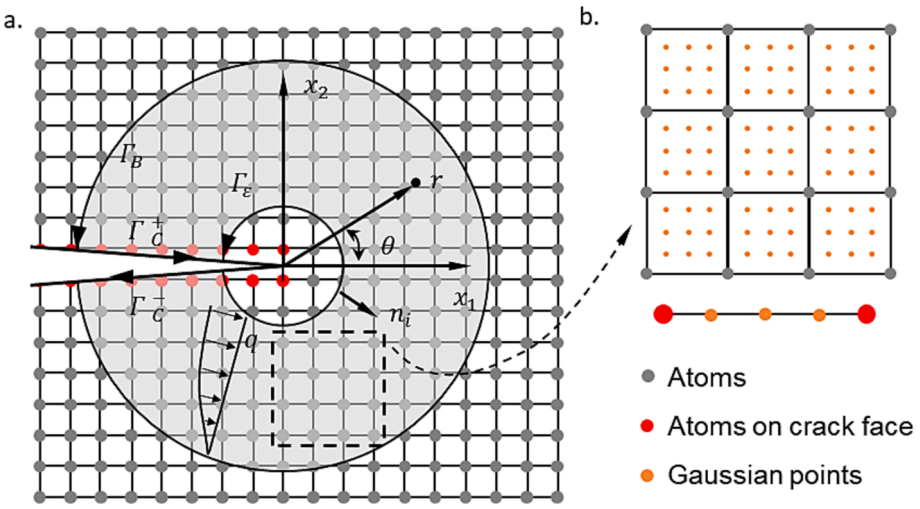


Fig. 1. (a) Local coordinates in the vicinity of the crack tip for J_{AFM}. (b) Atomic structure is meshed by linear quadrilateral isoparametric elements, and crack face is meshed by one-dimensional linear elements. Second-order Gaussian integration points are arranged inside the elements.

converted into an equivalent domain integral (EDI) to maintain high precision. In order to achieve the EDI form, the first step is to convert the integral contour Γ_ε in Eq. (1) into a closed contour Γ_0 surround the crack tip as illustrated in Fig. 1a. Γ_0 is defined as $\Gamma_0 = \Gamma_B^- + \Gamma_\varepsilon + \Gamma_C^+ + \Gamma_C^-$; Γ_C is the crack face; Γ_B^- and Γ_C^- are the opposite paths of Γ_B and Γ_C.

It should be noted that at the macroscale the traction-free condition is always substituted on the crack surface, i.e., $\sigma_{2j}(x_1, \pm 0) = 0$. However, the atomic forces often have non-zero values on the crack surface due to the long-range interaction between atoms. Thus, it is not rigorous to directly apply the hypothesis of CFM into the atomic-scale problems. In this case, we define the atomic J-integral based on the concept of AFM as

$$J_{\mathrm{AFM}} = \oint_{\Gamma_0} \left(W\delta_{1i} - \sigma_{ij}u_{j,1}\right) n_i q d\Gamma + J_{\mathrm{crk}} \tag{2}$$

where q is an arbitrary smooth weight function with the value of 1 on Γ_ε and 0 on Γ_B. The crack face integral J_{crk} is expressed as

$$J_{\mathrm{crk}} = \int_{\Gamma_C^+ + \Gamma_C^-} \left(W\delta_{1i} - \sigma_{ij}u_{j,1}\right) n_i q d\Gamma \tag{3}$$

Considering that the normal vector n_i on Γ_C is

$$n_1 = \begin{cases} 0 \ (\text{on } \Gamma_C^+) \\ 0 \ (\text{on } \Gamma_C^-) \end{cases}, \quad n_2 = \begin{cases} 1 \ (\text{on } \Gamma_C^+) \\ -1 \ (\text{on } \Gamma_C^-) \end{cases} \tag{4}$$

We can simplify J_{crk} as

$$J_{\mathrm{crk}} = \int_{\Gamma_C^-} \sigma_{2j}u_{j,1} q d\Gamma - \int_{\Gamma_C^+} \sigma_{2j}u_{j,1} q d\Gamma \tag{5}$$

Applying the divergence theorem to Eq. (2), we can obtain the EDI as

$$J_{\mathrm{AFM}} = \iint_A \left(W\delta_{1i} - \sigma_{ij}u_{j,1}\right) q_{,i} d\mathrm{A} + J_{\mathrm{crk}} \tag{6}$$

According to the existing research, CFM cannot provide reliable displacement and stress fields for J-integral when the specimen size decreases below the critical dimensional limits [30,31]. Meanwhile, how to accurately describe the constitutive near the crack tip with consideration of the large nonlinearity at fracture is quite challenging at such an atomic scale [39]. Thus, some improvements should be made to implement the J-integral calculations for nanostructures, including the discretization of the integral equation, the definition of stress and strain fields, etc.

In this work, AFM is used to provide real displacement and stress fields with consideration of atomic discreteness at the atomic scale. In the discrete system, the atomic displacements can be derived from the changing of coordinates for one certain atom before and after deformation. The atomic stress can be obtained from the Virial theorem, which gives the expression of the stress tensor as the function of atom coordinates and inter-atomic forces. The atomic stress tensor in the Cartesian coordinates at Atom α is expressed by

$$\sigma_{ij}^{\alpha} = \frac{1}{2V_0} \sum_{\beta \neq \alpha} r_i^{\alpha\beta} f_j^{\alpha\beta} \tag{7}$$

where V_0 is the volume per atom; $r_i^{\alpha\beta}$ is the distance between Atom α and Atom β in i direction; $f_j^{\alpha\beta}$ is the jth component of the interatomic force exerted on Atom α by Atom β [34].

Another problem appears in the definition of the displacement gradient and strain for discrete models at the atomic scale, since the strain was originally introduced for continuum media. For ideal crystals such as fcc and bcc, strain equals everywhere in the system, which is called the homogeneous strain system. Strain varies from location to location for materials that contain defects such as grain boundaries and cracks, which are called the nonhomogeneous strain systems. In such cases, the evaluation of local strain is often complicated due to the discontinuity of atoms. Especially for the cracked body, the displacement gradient in the vicinity of the crack tip changes significantly. In general cases, the local strain at Atom α is evaluated by taking an average of the displacements over its neighboring atoms by stain-displacement relationship [16]. However, the choice of which atoms should be included in the strain assessment is not rigorous. The cutoff radius can be changed to select the first or second neighboring atoms, while the calculations of displacement gradients are totally different when choosing different neighboring atoms.

An alternative approach is investigated referring to the idea of meshing in finite element method (FEM) to avoid calculating the displacement gradients directly at the atom. The atomic structures are meshed by linear quadrilateral isoparametric elements, as shown in Fig. 1b. The atoms along the crack face are meshed by one-dimensional linear elements to calculate the crack face integral. Displacement gradients and stress fields inside elements can be accurately calculated by shape functions. Since the integrand of EDI is continuous everywhere inside elements, Eq. (3) can be solved by the Gaussian integral method. The second-order Gaussian integration points are arranged inside elements, as shown in Fig. 1b. Regarding carbon atoms as "nodes" in FEM, the displacements and local stress fields of Gaussian points are obtained by element shape functions:

$$\begin{aligned}\boldsymbol{u} &= \left[N_1^e\mathbf{I}_{2\times2} \quad N_2^e\mathbf{I}_{2\times2} \quad N_3^e\mathbf{I}_{2\times2} \quad N_4^e\mathbf{I}_{2\times2}\right]\left[\boldsymbol{a}_1^e \quad \boldsymbol{a}_2^e \quad \boldsymbol{a}_3^e \quad \boldsymbol{a}_4^e\right]^{\mathrm{T}}\\ \boldsymbol{\sigma} &= \left[N_1^e\mathbf{I}_{3\times3} \quad N_2^e\mathbf{I}_{3\times3} \quad N_3^e\mathbf{I}_{3\times3} \quad N_4^e\mathbf{I}_{3\times3}\right]\left[\boldsymbol{\sigma}_1^e \quad \boldsymbol{\sigma}_2^e \quad \boldsymbol{\sigma}_3^e \quad \boldsymbol{\sigma}_4^e\right]^{\mathrm{T}}\end{aligned} \tag{8}$$

where $\mathbf{I}_{2\times2}$ and $\mathbf{I}_{3\times3}$ are the second-order and third-order identity matrices; $N_i^e(i=1,2,3,4)$ are the shape functions of linear quadrilateral isoparametric element; $(\boldsymbol{u},\boldsymbol{\sigma})$ and $(\boldsymbol{a}_i^e,\boldsymbol{\sigma}_i^e)$ are displacement and stress vectors of Gaussian points and atoms of the same element, respectively. $\boldsymbol{u}$, $\boldsymbol{\sigma}$, $\boldsymbol{a}_i^e$ and $\boldsymbol{\sigma}_i^e$ are expressed as

$$\begin{cases}\boldsymbol{u} = \begin{bmatrix}u\\ v\end{bmatrix},\ \boldsymbol{a}_i^e = \begin{bmatrix}u_i & v_i\end{bmatrix}\\ \boldsymbol{\sigma} = \begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{12}\end{bmatrix},\ \boldsymbol{\sigma}_i^e = \begin{bmatrix}\sigma_{11}^i & \sigma_{22}^i & \sigma_{12}^i\end{bmatrix}\end{cases} \tag{9}$$

where subscript i denotes the atom number in the element. The strain $\boldsymbol{\varepsilon}$ is then obtained from the geometric equation as

$$\boldsymbol{\varepsilon} = \begin{bmatrix}\frac{\partial}{\partial x} & 0\\ 0 & \frac{\partial}{\partial y}\\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x}\end{bmatrix}\boldsymbol{u} \tag{10}$$

3. Material and models

In this work, we take the graphene nanoribbons (GNRs) as a representative of 2D brittle materials with periodic hexagonal honeycomb structure [38,25,14,44,22,37]. A large number of theoretical studies on the brittle fracture of graphene have been done and CFM has been verified to be applicable in characterizing the fracture behavior of one-atom-thick GNRs [41,40,46,13,29,36,4,47,2].

3.1. Atomistic model

The calculation of J_{AFM} is based on the molecular statics (MS) simulation of quasi-static tensile tests for cracked GNRs, using the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [24]. The AIREBO potential (Adaptive Intermolecular Reactive Empirical Bond Order Potential) [34]is used to characterize the sp^2 bond between carbon atoms. The influence of kinetic energy is not considered during the whole process.

A single-edge-cracked specimen is introduced and stretched along the armchair direction, as shown in Fig. 2a. The geometric relationship of length H, width W, and crack length a is $H = 2W$, $W = 3a$. The thickness of monolayer graphene is regarded as 0.335 nm [23], which is the average interlayer distance of graphite. The lattice constant is 0.14 nm. Dimension along the length (armchair) direction is fixed by the periodic boundary condition. Considering that the unit cells of GNRs possess a hexagonal honeycomb structure, it is difficult to directly build an isoparametric element with six atoms as nodes. Thus, the graphene hexagonal cell is divided into two quadrilateral isoparametric elements instead, as shown in Fig. 2b. Nine Gaussian points are set for each element.

3.2. FEM model

The finite element analysis is performed in the following analysis to calculate the traditional J-integral as a comparison with J_{AFM} for the discrete atomic system. The FEM model is given in Fig. 2c. The dimensions of the FEM model are the same as those of MS simulation. The bottom boundary is fixed, and the critical displacement d obtained from atomic simulation is applied to the upper boundary. Ramberg-Osgood equation is assumed to describe the nonlinear elastic behavior in the uniaxial tension tests for GNRs, i.e.,

$$E\varepsilon = \sigma + \alpha_y\left(\frac{\sigma}{\sigma_0}\right)^{n-1} \tag{11}$$

The nonlinear elastic parameters for graphene are given in Table 1 and the detailed process to obtain these constants can refer to our previous work [15].

Table 1
Material properties of graphene in armchair direction [15].

Material property	Value
Young's modulus E (GPa)	968
Poisson's ratio ν	0.18
Yield stress σ_0 (GPa)	51.5
Work hardening coefficient n	4.9
Yield offset α_y	0.036

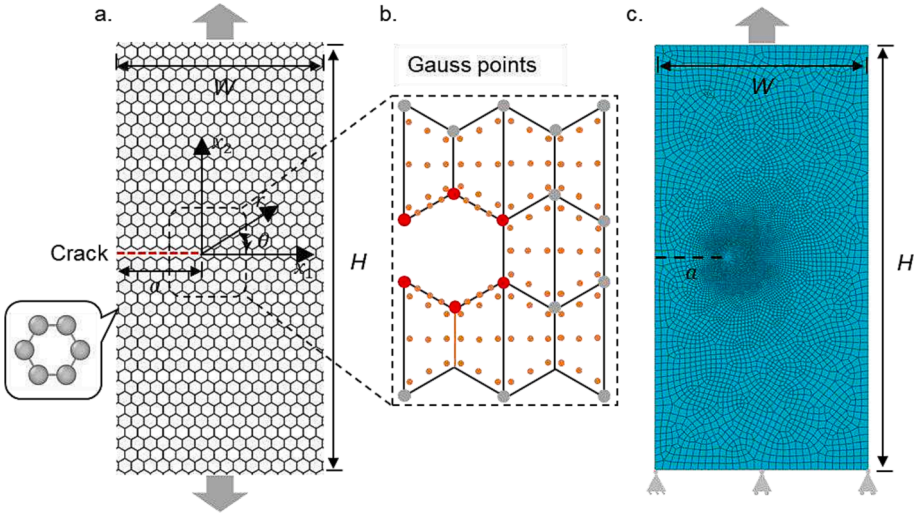


Fig. 2. (a) Geometry and loading conditions of a pre-crack GNR specimen for tensile tests. The crack face is along the zigzag direction. (b) The graphene hexagonal cells are divided into quadrilateral isoparametric elements. (c) Continuum model of cracked nano-specimens for FEM.

4. Results and discussion

The global stress–strain (σ^{global}-ε^{global}) relation of the pristine GNR demonstrates that graphene possesses eminent nonlinear elasticity at the tensile strain larger than 0.06 during the uniaxial tensile test, as shown in Fig. 3a. For cracked nano-sized GNRs, the global mechanical responses (σ^{global}-ε^{global} relations) of specimens with different sizes ($a = 0.7 \sim 14.0$ nm) are shown in Fig. 3b. It can be observed that the cracked GNRs can withstand the greater deformation at fracture when the specimen size decreases. Since the critical ε^{global} is always less than 0.06, the nonlinearity of the entire specimen is attributed to the local nonlinear deformation caused by the singular stress field in the vicinity of the crack tip [31]. With decrease of the specimen size, the shape of the σ^{global}-ε^{global} curves changes and represents stronger nonlinearity. In particular, the σ^{global}-ε^{global} curve is significantly bent when the crack length is reduced to $a = 0.7$ nm and 1.4 nm. Thus, the nonlinearity is of vital importance during the nano-fracture analysis and must be considered. Then, from the perspective of micro mechanics behavior, the fracture starts with the first bond breaking at the crack tip, and complete failure of the specimen proceeds subsequently and rapidly along the zigzag direction, resulting in two fractured pieces, as shown in Fig. 3c. Therefore, the failure of GNRs in this work is a purely brittle and mode-I fracture.

4.1. Verification of J_{AFM} method

In this section, two kinds of methods for calculating the J-integral, i.e., traditional FEM and stress-based method, are used to verify the accuracy of J_{AFM} in the following paragraph, and a GNR specimen with a crack ($a = 14.0$ nm) is modeled.

In the first part, the traditional FEM is used to calculate the stress and displacement fields of the cracked GNRs, and the J-integral, characterized by J_{FEM}, is obtained according to the EDI form of J-integral expressed by Eq. (3). During the uniaxial tension process in MS simulation, we extract the atomic configurations at four typical moments when ε^{global} along length direction reaches 0.5 %, 0.8 %, 1.0 % and the critical value ε_C^{global} (3.3 %), respectively. And the same displacement loads are applied to FEM models to calculate J_{FEM} at these four moments, respectively. Comparison between J_{AFM} and J_{FEM} is given in Table 2, which shows that the relative errors for all cases are within 3.5 %. At the fracture moment, the critical J-integral (J_{AFM}^F) represents the fracture toughness J_C of graphene along the zigzag direction. And the value of J_{AFM}^F agrees well with the result given by atomic energy release rate method, i.e., $J_C = G_{AFM}^C = 12.0$ J/m^2 [15]. Excellent agreement demonstrates that the J_{AFM} method is valid for the nonlinear elastic fracture problem of graphene at the nanoscale.

The stress-based method is used to calculate the J-integral directly, characterized by J_{MS}, through the analysis of the slope of the singular stress field ($\log\sigma_{yy}$-$\log r$, where r is the distance from the crack tip) in the vicinity of the crack tip. In practical research, a stress-based method is a direct and reliable way to estimate the value of the J-integral. Displacement of $\varepsilon^{global} = 1.0\%$ is applied to the specimen in this example. According to NLFM, the stress intensively concentrates near the crack tip and forms a singular field proportional to $r^{-1/(n+1)}$, i.e., the HRR (Hutchinson-Rice-Rosengren) singular stress field, which is expressed as [8]

$$\sigma_{ij}(r) = \left(\frac{J}{r}\right)^{\frac{1}{n+1}} \widetilde{\sigma}_{ij}(\theta, n) \quad (i,j = x,y) \tag{12}$$

where $n = 4.9$ is work hardening coefficient listed in Table 1; $\widetilde{\sigma}_{ij}(\theta, n)$ is

Table 2
J-integral for a GNR specimen with a crack length of $a = 14.0$ nm. Global strain ε^{global} takes 0.5 %, 0.8 %, 1.0 % and the critical value of 3.3 %, respectively.

ε^{global}	J_{AFM} (J/m^2)	J_{FEM} (J/m^2)	$error = \frac{\lvert J_{FEM} - J_{AFM}\rvert}{J_{FEM}}$
0.5 %	0.608	0.589	3.2 %
0.8 %	1.842	1.781	3.4 %
1.0 %	3.023	2.946	2.6 %
3.3 %	12.087	11.898	1.6 %

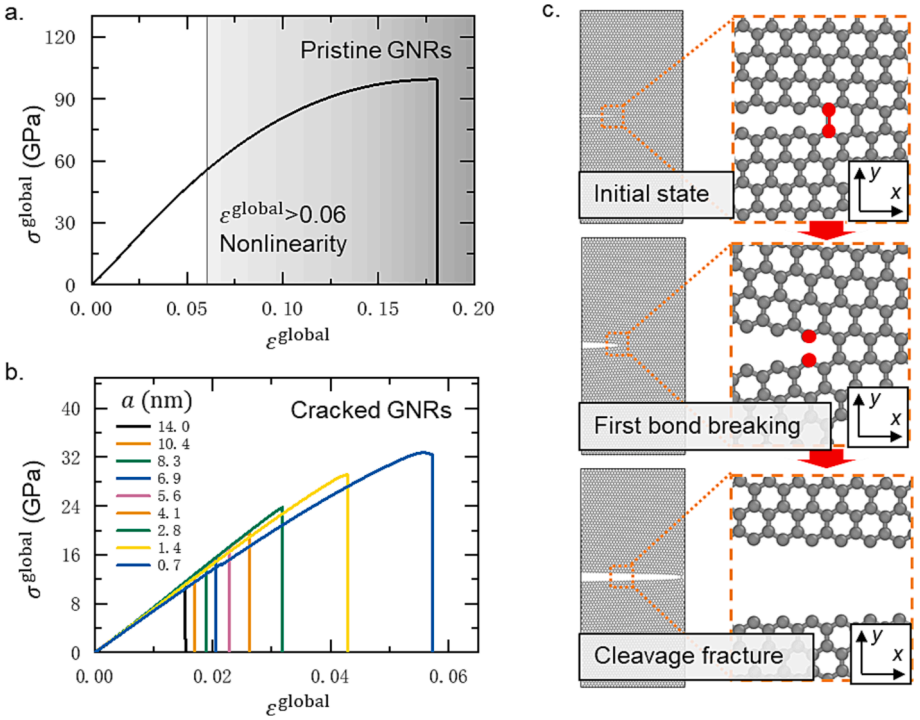


Fig. 3. (a) σ^{global}-ε^{global} curve along the armchair direction of the pristine GNR. (b) σ^{global}-ε^{global} curves of cracked GNRs with different dimensions of $a = 0.7 \sim 14.0$ nm. (c) Atomic configurations of the GNR specimen at the onset of crack propagation and after crack propagation.

the dimensionless function of n and θ. The J-integral for an atomic structure can be directly obtained by the stress-based method following Eq. (12), instead of calculating the domain integral expressed by Eq. (6). As shown in Fig. 4a, the normal stress σ_{yy} extracted from the MS simulation results well exhibits the HRR singularity. The values of J_{MS} obtained by Eq. (12) for the atoms near the crack tip are 2.577 J/m^2, 2.964 J/m^2, 2.955 J/m^2, 3.060 J/m^2 and 3.064 J/m^2, as shown in Fig. 4b. Results indicate that $J_{MS} = 2.964 \sim 3.064\,\mathrm{J/m^2}$ agree well with $J_{AFM} = 3.023\,\mathrm{J/m^2}$ and this verifies the accuracy of J_{AFM} again. However, it should be noticed that $J_{MS} = 2.577\ \mathrm{J/m^2}$ at the atom nearest the crack tip, which is quite different from $J_{AFM} = 3.023\,\mathrm{J/m^2}$. This demonstrates that the normal stress obtained from MS simulation satisfies the HRR singularity no longer within such an atomic-scale region near the crack tip. Moreover, it is inevitable that J_{MS} slightly varies for different atoms, indicating that the stress-based method can only provide an approximate range of J_{MS}, but fails to calculate the exact value. Therefore, we can claim that, comparing with the stress-based method, the J_{AFM} method is more accurate and reliable when calculating the J-integral at the nanoscale.

4.2. Path-independency of J_{AFM}

In this case, we will discuss the path-independency of J_{AFM} when the size of the integral region changes. Here, the path-independency of J_{AFM} implies that J_{AFM} is not influenced when the integral region varies. The specimen with crack length of $a = 21.2\,\mathrm{nm}$ under critical deformation ($J_{AFM} = J^F_{AFM}$) is taken as the example in this section. As shown in Fig. 5a, we apply the circle integral region with dimension characterized by R_I. During the tests, R_I varies from 0.5 nm to 12.0 nm to control the size of integral region. The weight function q used in the J-integral of EDI form at Point P is given as

$$q(r_P) = \frac{R_I - r_P}{R_I} \tag{13}$$

where r_P is the distance between Point P and the crack tip. The results given in Fig. 5b demonstrate that J^F_{AFM} remains independent of the integral region as long as R_I is large enough, i.e., $J^F_{AFM} = J_C$ when $R_I \geq 2.0\,\mathrm{nm}$. However, when R_I becomes smaller, J^F_{AFM} begins to deviate from J_C, as shown in Fig. 6a, which indicates that the J_{AFM} method loses its path-independency.

Then, we will find out the reasons of the loss of path-independency. The specimens with the dimensions of $a = 2.8\,\mathrm{nm}$ and $a = 14.0\,\mathrm{nm}$ are calculated here to discuss the influence brought by the specimen size. In addition, the square shaped integral domain is used here to compare

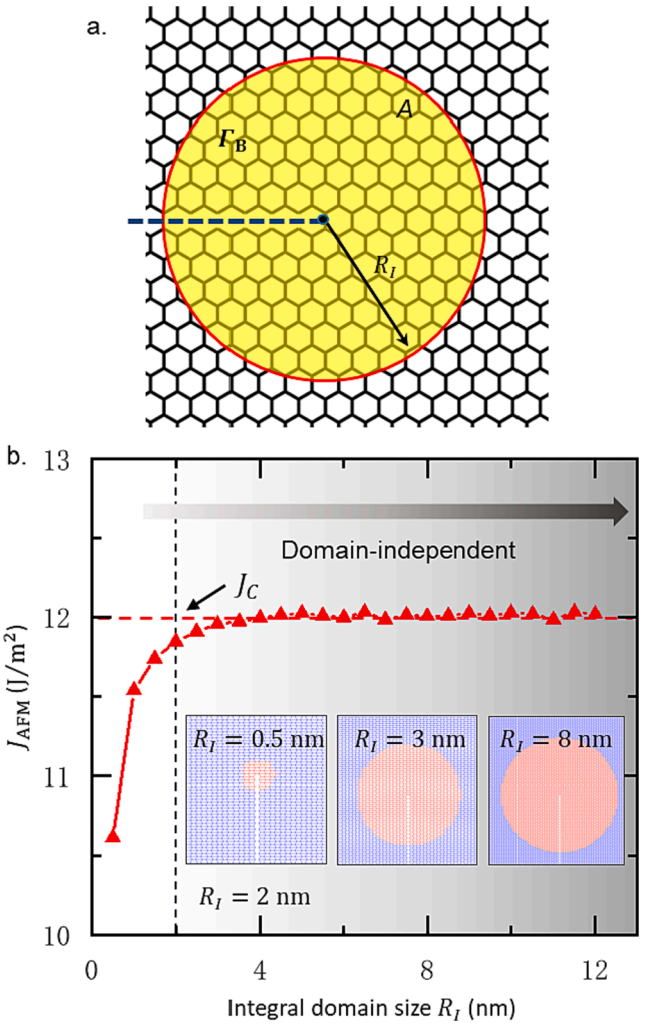


Fig. 5. (a) Circle shaped integral path of J_{AFM}. (b) J_{AFM} for different integral domain sizes $R_I = 0.5 \sim 12.0\,\mathrm{nm}$. The specimen with crack length of $a = 21.2\,\mathrm{nm}$ under critical deformation is used in this example.

with the circle case, as shown in Fig. 6a. The weight function q at any point P (x_P, y_P) in the area A for the square shaped case is given as

$$q = \frac{1}{r^4}\left(r^2 - x_P^2\right)\left(r^2 - y_P^2\right) \tag{14}$$

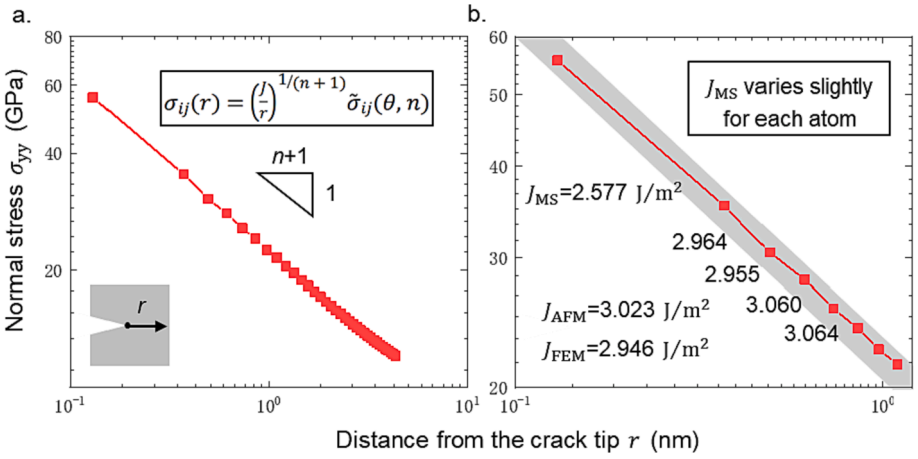


Fig. 4. (a) Atomic normal stress as a function of the distance from the crack tip given by MS simulation. It exhibits the HRR singularity of $r^{-1/n+1}$. (b) Atomic normal stress within the region of 1 nm from the crack tip. J_{MS} for the atoms near the crack tip are given in the figure.

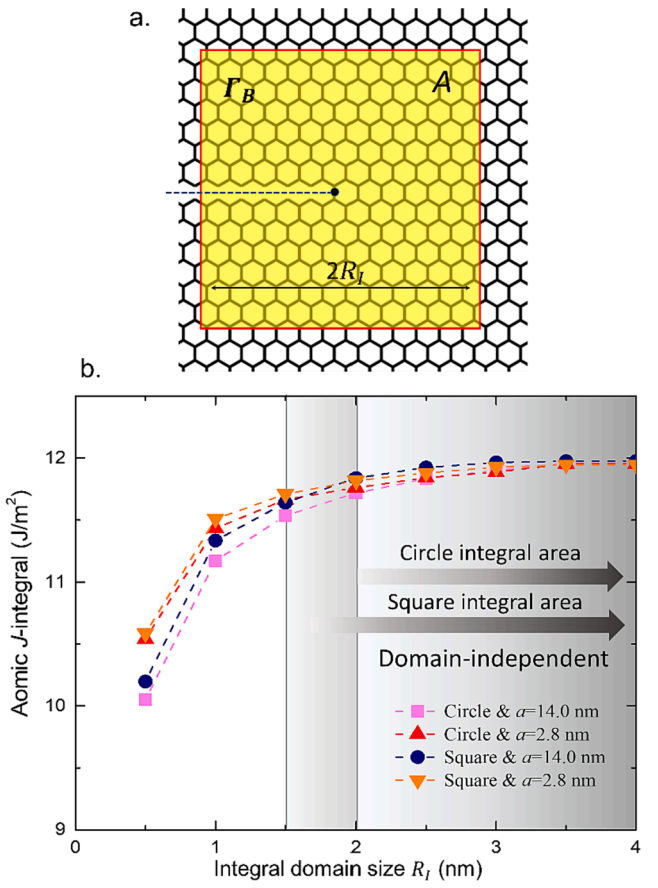


Fig. 6. (a) Square shaped integral path of J_{AFM}. (b) J_{AFM} for different integral domain sizes $R_I = 0.5 \sim 4.0$ nm. The cases of $a = 2.8$ nm and $a = 14.0$ nm are considered here.

The results given in Fig. 6b demonstrate that J_{AFM} is path-independent when $R_I \geq 1.5$ nm for square case and $R_I \geq 2.0$ nm for circle case. Therefore, the path-dependency appears when selecting different kinds of the integral shapes and weight functions. On the other hand, the variation of J_{AFM} with R_I is almost independent of the specimen size a. Although $a = 14.0$ nm is 4 times larger than $a = 2.8$ nm, J_{AFM} remains almost unchanged when selecting the same R_I, which indicates the path-independency is governed by the localized area near the crack tip.

In our opinion, the loss of path-independency is attributed to the atomic-scale geometric nonlinearity near the crack tip. In the prove process of the conservation of J-integral, the infinitesimal deformation is a basic assumption adopted. In NLFM, the strain defined by Eq. (10) is called the Cauchy strain, which is useful only for small displacement gradients. And we also apply the Cauchy strain during the calculation of J_{AFM}. However, finite deformation might exist near the crack tip in AFM simulation, leading to the infinitesimal deformation assumption no longer applicable for nano-scale brittle fracture issues. In order to verify this view, Green strain is introduced here to describe the finite deformation around the crack tip. The components of Green strain tensor in Cartesian coordinates are

$$E_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i} + u_{k,i}u_{k,j}\right) \ (i,j,k = 1,2) \tag{15}$$

Applying the Green strain and Cauchy strain to analyze the displacements of atoms near the crack tip, respectively. The normal strain fields ε_{yy} are given in Fig. 7. Results demonstrates that within the range of 0.12 nm around the crack tip, ε_{yy} given by the Green strain is significantly greater than the Cauchy strain, which indicates that the infinitesimal deformation assumption has failed in this area. Here, $\Lambda_{\mathrm{GN}} =$

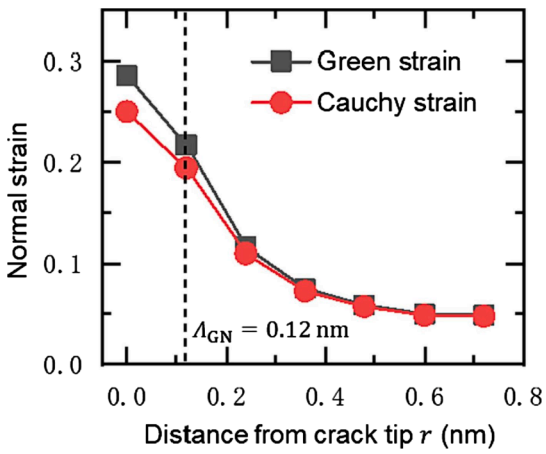


Fig. 7. Normal strain in Green strain and Cauchy strain form. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

0.12 nm is defined as the size of the geometric nonlinear area. When R_I is large enough, i.e., $R_I \gg \Lambda_{\mathrm{GN}}$, the influence brought by the geometric nonlinearity is significantly small that can be ignored. As R_I decreasing to 2.0 nm or less, R_I and Λ_{GN} almost reach the same order of magnitude, and the geometric nonlinearity will have an obvious impact. In such case, the infinitesimal deformation assumption breaks down at the atomic scale, and this will finally lead to the loss of path-independency for J_{AFM}. On the other hand, the constitutive of the material should be linear/nonlinear elastic to make the J-integral path-independent, within the framework of CFM. However, MS provides the real stress field, which is smaller than the singular stress described by NLFM. Therefore, the atomic stress shows much stronger nonlinearity near the crack tip and does not satisfy the HRR singularity, as shown in Fig. 4b. This real nonlinearity also contributes to the loss of the path-independency.

In above, the J_{AFM} method constructed by AFM is reliable as long as the integral region around the crack tip is large enough to guarantee the integral path-independency.

4.3. Dimension-independency of J_{AFM}

In this section, we will discuss the application of J_{AFM} method when dimension decreases to extremely small size at the nanoscale. The critical J-integral at fracture for different specimen sizes with the same geometric and boundary conditions are calculated by both FEM ($J^{\mathrm{F}}_{\mathrm{FEM}}$) and the atomic method ($J^{\mathrm{F}}_{\mathrm{AFM}}$). Fig. 8a plots the J-integral at fracture as a function of the crack length a for all tested GNRs. It shows that $J^{\mathrm{F}}_{\mathrm{AFM}}$ remains consistent with J_{C} as the specimen size varies from $a = 2.8$ nm to $a = 14.0$ nm. This demonstrates that J_{AFM} is dimension-independent at the atomic scale. On the other hand, $J^{\mathrm{F}}_{\mathrm{FEM}}$ is consistent with the fracture toughness J_{C} when $a > 6.9$ nm. This consistency evidently indicates that the NLFM criterion, i.e., the crack begins to propagate when J_{FEM} reaches the fracture toughness J_{C}, is applicable. However, when dimension further decreases below the critical length, i.e., $a^{\mathrm{C}}_{\mathrm{NL}} = 6.9$ nm, $J^{\mathrm{F}}_{\mathrm{FEM}}$ begins to deviate from the fracture toughness J_{C}, which implies that NLFM fails to describe the fracture in GNRs. In another word, the CFM-based J-integral criterion breaks down at the atomic scale when $a < a^{\mathrm{C}}_{\mathrm{NL}}$.

Extracting the normal stress σ_{yy} in the vicinity of the crack tip, Fig. 8b indicates that the stress given by AFM remains lower than the ideal strength σ_{IS} ($\sigma_{\mathrm{IS}} = 99.8$ GPa). Here, σ_{IS} represents the fracture strength of a perfect graphene sheet under uniform deformation, which is equivalent to the strength of sp^2 C—C bond. As the specimen size approaches the macroscale ($a = 10.4$ nm), σ_{yy} calculated by FEM near the crack-tip is relatively well approximated by continuum assumption, as

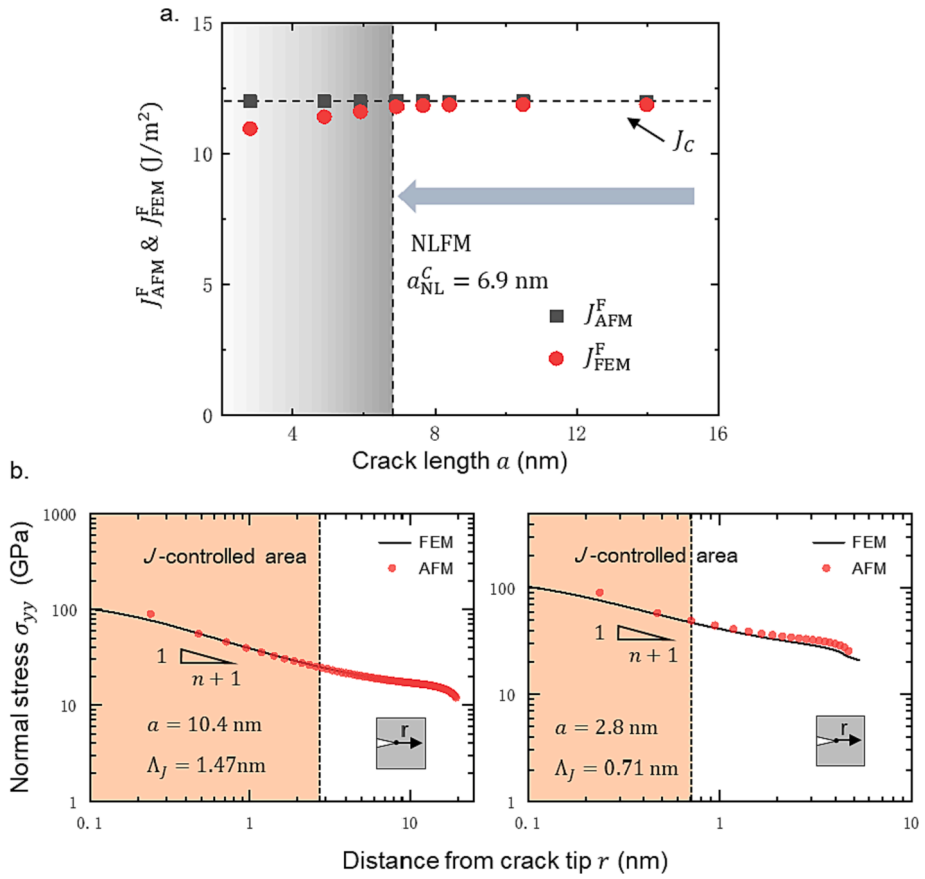


Fig. 8. (a) Fracture J-integral of graphene specimens with different dimensions ($a = 2.8 \sim 14.0$ nm) based on NLFM and the J_{AFM} method. (b) Normal stress at the critical fracture moment as a function of the distance from the crack tip.

shown in Fig. 8b. It exhibits HRR singularity in its J-controlled area within the range of $\Lambda_J = 1.47$ nm. However, when specimen size decreases to the atomic scale ($a = 2.8$ nm), results show that σ_{yy} obtained by continuum assumption deviates from the actual atomic one. And the J-controlled area size shrinks down to $\Lambda_J = 0.71$ nm at this scale, which is almost the same order of magnitude as the distance between carbon atoms. In such atomic scale, the discreteness of the atoms near the crack tip become much stronger. But the NLFM-based singularity cannot take into account the crack-tip atomic features, such as structural rearrangement, and this ultimately leads to the breakdown of NLFM. In contrast, the stress fields and displacements used in the J_{AFM} method are provided by atomic models without any continuum assumption, and this ensures the dimension-independency of J_{AFM} at the nanoscale. Therefore, the J_{AFM} method proposed in this study has a great advantage in solving the atomic-scale fracture problems compared with the traditional CFM methods.

5. Conclusions

In conclusion, this paper proposes an AFM-based J-integral (J_{AFM}) for brittle nonlinear fracture at the nanoscale. The creative points can be summarized as follows:

1) The line integral along the crack surface is considered in the J-integral for atomic structures.
2) A critical difficulty in calculating the J-integral for a discrete system, i.e., how to describe the local displacement gradient of atoms, is overcome through meshing the atomic structure by isoparametric elements.
3) The applicability and path-independency of the J_{AFM} method are verified. The geometric nonlinearity near the crack tip is discussed by introducing the Green strain. It reveals the failure of the infinitesimal deformation assumption when the integral region shrinks down to the atomic scale.
4) The breakdown of NLFM at the nanoscale is investigated, and the lower dimensional limit of NLFM for brittle fracture in Zigzag direction of graphene nanoribbons (GNRs) is found, i.e., $a^{C}_{NL} = 6.9$ nm.
5) The dimension-independency of the J_{AFM} method is confirmed. It demonstrates that J_{AFM} is valid at all scales even below the critical dimensional limit of NLFM.

CRediT authorship contribution statement

Pengfei Jia: Formal analysis, Investigation, Software, Visualization, Writing – original draft. **Kai Huang:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Hongjun Yu:** Validation, Writing – review & editing. **Takahiro Shimada:** Writing – review & editing. **Licheng Guo:** Resources, Supervision, Writing – review & editing. **Takayuki Kitamura:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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